MILITARY SPECIFICATION

ROCKET MOTOR/BULKHEAD ASSEMBLY
M753 8 INCH PROJECTILE

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for rocket motor/bulkhead assemblies used on M753, 8 Inch Projectiles (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-H-1168	Helium, Technical
QQ-A-1876	Aluminum Foil

MILITARY

MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-C-16173	Corrosion Preventive Compound, Solvent Cutback, Cold- Application

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

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STANDARDS

FEDERAL

FED-STD-209	Clean Room and Work Station Requirements, Controlled Environment

MILITARY

MIL-STD-453	Radiographic Inspection
MIL-STD-1167	Ammunition Data Cards

2.1.2 Other Government documents, drawings and publications. The following other Government documents form a part of this specification to the extent specified herein.

DRAWINGS (see 6.11)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9287988	Adhesive
9299053	Motor Body Bonding Assembly, M753
9299056	Motor Body Delay Assembly, M753
9299057	Rocket Motor, 8 Inch, M753, Live
9299058	Rocket Motor/Bulkhead Assembly, 8 Inch, M753
9311169	Packing and Marking for Drum, Metal Shipping for Rocket Motor, 8 Inch
9328029	Rocket Test Vehicles, PXR6275
9328448	Insulator, Bulkhead
9331235	Motor Body, Match Drill
9331658	Motor Body and Warhead Match Drill Assembly
9331659	Aft/Fwd Warhead Assembly
9331661	Aft Warhead/Fwd Warhead/Insulator Bonding Assembly
9335593	Compound, Silicone Rubber
9335594	Primer, Bonding
9335595	Insulator Bonding Procedure
9340549	Procedure for Obturator Installation, M753
9340585	Motor Body Bonding Assembly Procedure
9340586	Motor Body Delay Assembly Procedure
9340656	Bulkhead Bonded Subassembly
9341458	Rocket Motor/Bulkhead Assembly Procedure

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9349957	Vapor Barrier, Bonding Procedure
9349958	Vapor Barrier
9347149	Rocket Motor/Bulkhead Assembly, Type 6B, Live
11828813	Adhesive, Polyamide Epoxy

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein, the text of this specification shall take precendence.

3. REQUIREMENTS

3.1 Parts and materials. The contractor shall provide objective evidence (see 6.6) that parts and materials comply with applicable drawing (dwg.) and specification requirements.

3.2 Assembly. Assemblies shall comply with all requirements of dwg. 9299058 (see 6.10), with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the dwg. and herein.

3.3 Clean room. Assembly shall be performed in an approved clean room meeting the requirements of FED-STD-209 and the guidelines specified herein (see 6.7).

3.4 Inspection records and data cards. Inspection records of all inspections performed on individual rocket motor/bulkheads and ammunition data cards prepared in accordance with MIL-STD-1167 shall be provided. One copy of the ammunition data cards shall be inclosed in the 9311169 shipping container.

3.5 Pin extraction force. Neither the retaining pins nor the retaining rings of rocket motor/bulkhead assemblies shall be extractable from pin holes by a pin axial force of 1500 + 50 pounds (lbs). Testing shall be performed in accordance with 4.5.4.

3.6 Leak rate. The leak rate of the rocket motor/bulkhead joint and pin holes shall not exceed 1×10^{-6} atm-cc/sec (air equivalent). Testing shall be performed in accordance with 4.5.5.

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3.7 Delay removal force. The minimum axial force required to remove the delay assembly from the rocket motor/bulkhead assembly shall be 400 lbs. Testing shall be performed in accordance with 4.5.6.

3.8 Proving ground tests. Rocket motors shall comply with the requirements specified herein when assembled on a rocket test vehicle (RTV) furnished by the acquistion agency and fired at a Government proving ground. Testing shall be performed in accordance with 4.5.7.

3.8.1 Metal part integrity. There shall be no premature burst or indication of metal part breakup, collapse or separation in the gun bore or in flight.

3.8.2 Motor function. With the rocket-off cap removed (rocket-on mode) the rocket motor shall not ignite in the gun bore but shall ignite and fire in flight.

3.9 Government furnished property. Unless otherwise indicated in the contract, the following items shall be Government furnished, as required.

Quantity	Description	Drawing
One per rocket motor/bulkhead assembly	Match drilled motor body/ bulkhead shipping assembly	AY236433 (SLL)(REF)
20	Aft/Fwd Warhead Assemblies for PXR6275-2 first article proving ground test vehicles (includes match drilled motor body/aft whd. without bulkhead insulators)	9331659
20	Bulkhead insulators (for assembly on first article Aft/Fwd Warhead Assembly of PXR6275-2 proving ground test vehicles in accordance with dwg. 9331661)	9328448

3.10 First article inspection. This specification contains technical provisions for first article inspection (see 4.3). Requirements for submission of first article samples shall be as specified in the contract.

3.11 Workmanship. The requirements for workmanship shall be as specified on applicable dwgs. and as specified herein.

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3.11.1 Cleaning. Parts and assemblies shall be clean and free of chips, dirt, grease, rust, corrosion and other foreign matter prior to painting. Cleaning methods shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

3.11.2 Painting. Painting shall comply with all applicable requirements of applicable dwgs. Exterior coatings shall be continuous with the possible exception of a few light touched up scratches not exposing base materials. Drying time shall be in accordance with dwg. requirements when specified and all paint shall be dry to the touch before packing for shipment.

3.11.3 Marking. Marking shall be in accordance with applicable dwgs. and shall be neat, complete, sharply defined and in the correct location.

3.11.4 Rotating bands. Care shall be exercised at all times to assure that the rotating band does not become nicked or damaged. Where it is necessary to remove the grommet assembly during assembly operations, the rotating band shall be examined for gross damage prior to reassembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3)
2. Quality conformance inspection (see 4.4)

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4.3 First article inspection.

4.3.1 Submission. Where first article inspection is required (see 3.10), the contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Quantity	Description	Drawing
20	PXR6275-2 Rocket Test Vehicles with rocket motors assembled in stages (see 4.3.2.1.1)	9328029
2	Motor Body Delay Assemblies for delay removal force test.	9299056

4.3.2 Inspections to be performed. Inspection shall be performed in accordance with MIL-A-48078 and the first article inspection table herein subject to the conditions of 4.3.2.1.1.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.2.1	Rocket Motor/Bulkhead Assembly, 8 Inch, M753	1 OF 1	9299058
			NEXT HIGHER ASSEMBLY: M753 Projectile

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Objective evidence for materials and components	-		3.1	4.4.2.1
	Classification of defects and tests (see 4.3.2.1.1)	20			
	1. Bulkhead Bonded Subassembly			3.2, 3.11	4.4.2.2
	2. Motor Body Bonding Assembly			3.2, 3.11	4.4.2.3
	3. Motor Body Delay Assembly			3.2, 3.11	4.4.2.4
	4. Rocket Motor, 8 Inch (Live)			3.2, 3.11	4.4.2.5
	5. Rocket Motor/RTV Assembly			3.2, 3.5, 3.6, 3.11	4.4.2.6
	6. Proving ground tests (note 1)			3.8	4.5.7
	Delay assembly removal force (note 1)	2		3.7	4.5.6

NOTES: 1. Destructive tests

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4.3.2.1.1 Order of inspections. The first article PXR6275-2 Test Projectiles for proving ground tests shall be assembled following the same assembly and inspection procedures as normally used in war reserve (WR), rocket motor/bulkhead assemblies except that the Gov't furnished PXR6275-2 Match Drill Motor Body/Aft Warheads shall be used in lieu of WR Match Drill Motor Body/Bulkhead Assemblies and the Gov't furnished PXR6275-2 Aft Warhead shall apply in lieu of the WR Bulkhead. The first article inspection shall be performed in stages as follows. Twenty Gov't furnished warhead insulators shall be bonded onto Gov't furnished aft warhead assemblies in accordance with dwgs. 9340656 and 9335595. Following preliminary Gov't acceptance of these items in accordance with 4.4.2.2, the twenty Gov't furnished match drill motor bodies and WR rocket motor components shall be assembled into motor body bonding assembies in accordance with dwgs. 9299053 and 9340585. Following preliminary Gov't acceptance of these items in accordance with the inspection requirements of 4.4.2.3, the motor body bonding assemblies shall be further assembled into motor body delay assemblies in accordance with dwgs. 9299056 and 9340586. Following preliminary Gov't acceptance of these items in accordance with 4.4.2.4, the motor body delay assemblies shall be further assembled into rocket motors in accordance with dwg. 9299057. Following preliminary Gov't acceptance of these items in accordance with the inspection requirements of 4.4.2.5, the rocket motors shall be assembled into PXR6275-2 Rocket Test Vehicles in accordance with dwg. 9328029. Inspection of these test vehicles shall be performed in accordance with 4.4.2.6 except that the applicable inspections normally performed on rocket motor/bulkhead assemblies shall be performed on the rocket test vehicles. Following Gov't acceptance of the rocket test vehicles in accordance with 4.4.2.6, the vehicles shall be forwarded to a Gov't proving ground for proving ground tests in accordance with instructions from the cognizant ARRADCOM element (see 6.4).

4.3.3 Inspection equipment. The inspections and tests specified in 4.3.2 shall be accomplished with gages and test equipment specified in 4.4.4.

4.3.4 Rejection. See MIL-A-48078.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall conform with the lot formation provisions of MIL-A-48078. In

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addition, each inspection lot of loaded rocket motor/bulkhead assemblies shall contain:

a. Motor body metal parts of one lot interfix number from one manufacturer.
b. Bulkhead insulations of one lot interfix number from one manufacturer
c. Delay assemblies of one lot interfix number from one manufacturer
d. Propellant grain of one lot interfix number from one manufacturer.

4.4.2 Examinations. See MIL-A-48078 and the Quality Conformance Inspection tables herein.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE			SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Materials and Components				See Below
					NEXT HIGHER ASSEMBLY
					See Below
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101.	Objective evidence is required for the following items in accordance with applicable dwgs. and specifications			3.1	Visual

Identification	Dwg. No.	Ref. on Dwg.
Adhesive, Polyamide Epoxy	11828813	9340585
Adhesive	9287988	9340585
Corrosion Preventive Compound	MIL-C-16173	9341458
Silicone Rubber	9335593	9335595
Primer Bonding	9335594	9335595
Vapor Barrier	QQ-A-1876	9349958

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Bulkhead Bonded Subassembly	1 OF 4	9340656, 9349957, 9335595
			NEXT HIGHER ASSEMBLY: 9299058 (see 6.10)

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101.	Objective evidence for bulkhead insulator and vapor barrier		-	3.1	Visual
102.	Improper abrasion of bulkhead insulator bonding surface		100%	3.2	Visual
103.	Improper bulkhead insulator fit into bulkhead recess		100%	3.2	Visual
104.	Certification (or recertification) for primer and silicone rubber compound		-	3.2, 6.5	Visual
105.	Bulkhead insulator not clean and free of dirt, dust, grease or other foreign matter before bonding		100%	3.2	Visual
106.	Primer missing from insulator		100%	3.2	Visual
107.	Improper primer curing time		100%	3.2	Record
108.	Bulkhead bonding surface not clean and free of dirt, dust, grease or other foreign matter before bonding		100%	3.2	Visual
109.	Improper or missing extrusion of silicone rubber around edges for 360°		100%	3.2	Visual
110.	Improper bulkhead/insulator bond curing time		100%	3.2	Record

NOTES: 1. Bulkhead Bonded Subassemblies which are rejected may be reworked in accordance with applicable dwg. and approved procedures and reinspected for the applicable requirements herein.

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PARAGRAPH	TITLE		SHEET 2 OF 4		DRAWING NUMBER 9340656, 9349957, 9335595
4.4.2.2 (cont)	Bulkhead Bonded Subassembly				NEXT HIGHER ASSEMBLY 9299058 (see 6.10)
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
111.	Insulator not secured to bulkhead		100%	3.2	Manual
112.	Excessive silicone rubber not removed		100%	3.2	Visual
113.	Certification of insulator bond test specimens for sheer strength (see note 2)		-	3.2, 3.2, 6.5	Visual
114.	Infra-red inspection of bulkhead/ insulator bond		100%	3.2	4.5.2
115.	Vapor barrier improperly cleaned and prepared before bonding		100%	3.2	Visual
116.	Vapor barrier bonding surfaces improperly grit blasted before bonding		100%	3.2	Visual

NOTE 2. Five bond test coupons processed in the same manner as the bonds they represent shall be provided to the ARRADCOM Adhesives Laboratory (See 6.5) for each mix or batch of the applicable adhesive. The test coupons shall be positively identified with traceability to the serialized bulkhead bonded assemblies on which the mix or batch of adhesive was used and shall be identified on all applicable data cards.

NOTES:

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PARAGRAPH	TITLE		SHEET 3 OF 4		DRAWING NUMBER
4.4.2.2 (cont)	Bulkhead Bonded Subassembly				9340656, 9349957, 9335595
					NEXT HIGHER ASSEMBLY 9299058 (see 6.10)
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
117.	Vapor barrier bonding surfaces not clean and free of dirt, dust, grease or foreign matter before vapor barrier bonding		100%	3.2	Visual
118.	Vapor barrier improperly assembled or missing		100%	3.2	Visual
119.	Improper presence of vapor barrier bubbles		100%	3.2	Visual
120.	Excessive vapor barrier adhesive not removed		100%	3.2	Visual
121.	Improper vapor barrier adhesive curing time		100%	3.2	Record
122.	Certification of vapor barrier adhesive test specimens (see note 2)		-	6.5	Visual
	NOTE 2. Five bond test coupons processed in the same manner as the bonds they represent shall be provided to the ARRADCOM Adhesives Laboratory (See 6.5) for each mix or batch of the applicable adhesive. The test coupons shall be positively identified with traceability to the serialized bulkhead bonded assemblies on which the mix or batch of adhesive was used and shall be identified on all applicable data cards.				

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PARAGRAPH	TITLE		SHEET 4 OF 4		DRAWING NUMBER
4.4.2.2 (cont)	Bulkhead Bonded Subassembly				9340656, 9349957, 9335595
					NEXT HIGHER ASSEMBLY 9299058 (see 6.10)
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
123.	Vapor barrier acceptance stamp missing		100%	3.2	Visual
124.	Protective cap missing or improperly taped		100%	3.2	Visual
125.	Poor workmanship		100%	3.11	Visual

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Motor Body Bonding Assembly	1 OF 2	9299053 9340585
			NEXT HIGHER ASSEMBLY 9299056

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101.	Objective evidence for nozzle insert, grain support and aft cushion		-	3.1	Visual
102.	Motor body, grain support and aft cushion surfaces to be bonded not clean and free of rust, dirt, corrosion and foreign matter		100%	3.2	Visual
103.	Improper nozzle insert fit		100%	3.2	Gage
104.	Nozzle insert missing, cracked, damaged or improperly assembled		100%	3.2	Visual
105.	Nozzle insert not flush or below flush		100%	3.2	Manual
106.	Improper nozzle insert adhesive bond curing time		100%	3.2	Record
107.	Nozzle insert insecurely bonded		100%	3.2	Manual
108.	Grain support missing or improperly assembled		100%	3.2	Visual
109.	Improper grain support adhesive bond curing time		100%	3.2	Record
110.	Grain support insecurely bonded		100%	3.2	Manual
111.	Aft cushion missing or improper		100%	3.2	Visual
112.	Improper aft cushion adhesion bond curing time		100%	3.2	Record
113.	Aft cushion insecurely bonded		100%	3.2	Manual

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PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9299053 9340585
4.4.2.3 (cont)	Motor Body Bonding Assembly				NEXT HIGHER ASSEMBLY 9299056
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
114.	Excessive adhesive in motor body or nozzle insert cavity		100%	3.2	Visual
115.	Location of top of cushion		100%	3.2	Gage
116.	Poor workmanship		100%	3.11	Visual

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Motor Body Delay Assembly	1 OF 1	9299056 9340586
			NEXT HIGHER ASSEMBLY 9299057

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101.	Objective evidence for delay assembly, rocket-off cap assembly, cap seal, shear seal ring and preformed O-ring		---	3.1	Visual
102.	Components and delay assembly/motor body interface surfaces not clean and free of rust, dirt, grease and foreign matter		100%	3.2	Visual
103.	Delay assembly O-ring missing prior to delay assembly to the motor body		100%	3.2	Visual
104.	Chalked scribe mark showing orientation of delay columns missing		100%	3.2	Visual
105.	Rocket-off cap O-ring missing		100%	3.2	Visual
106.	Rocket-off cap seal missing		100%	3.2	Visual
107.	Improper rocket-off cap assembly torque		100%	3.2	Gage
108.	Missing or improperly assembled components		100%	3.2	X-Ray/4.5.1
109.	Poor workmanship		100%	3.11	Visual

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Rocket Motor, 8 Inch (Live)	1 OF 1	9299057
			NEXT HIGHER ASSEMBLY: 9299058 (see 6.10)

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101.	Objective evidence for obturator and propellant grain		---	3.1	Visual
102.	Obturator improperly assembled, missing or damaged (see dwg. 9340549)		100%	3.2	Visual
103.	Propellant grain improper, visibly damaged or cracked		100%	3.2	Visual
104.	Interface surfaces not clean and free of rust, dirt, grease and foreign matter		100%	3.2	Visual
105.	Inhibited end of grain not facing aft		100%	3.2	Visual
106.	Propellant grain improperly oriented with respect to monitoring port		100%	3.2	Visual
107.	Color band or patch missing or improper		100%	3.2	Visual
108.	Marking missing, improper or illegible		100%	3.2	Visual
109.	Poor workmanship		100%	3.11	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
4.4.2.6	Rocket Motor/Bulkhead Assembly				9299058 (see 6.10) 9341458
					NEXT HIGHER ASSEMBLY M753 PROJ
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101.	Objective evidence for metal C-Ring, forward cushion, retaining rings, monitoring port ball seal, self-locking setscrews and pins		---	3.1	Visual
102.	Rocket motor/bulkhead serial numbers don't match		100%	3.2	Visual
103.	Position of slot in propellant grain in relation to monitoring port improper (recheck)		100%	3.2	Visual
104.	Temperature and relative humidity of assembly room		100%	3.2	Record
105.	Improper forward cushion thickness		100%	3.2	Gage
106.	Forward cushion/bulkhead bonding subassembly bonding surfaces not clean and free of dirt and other foreign matter		100%	3.2	Visual
107.	Forward cushion missing or insecurely bonded		100%	3.2	Visual/Manual
108.	Motor body and bulkhead bonding subassembly seal surfaces not clean and free of burrs, grease, rust or other foreign matter		100%	3.2	Visual

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PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.6 (cont)	Rocket Motor/Bulkhead Assembly		2 OF 3	9299058 (see 6.10) 9341458
				NEXT HIGHER ASSEMBLY: M753 PROJ

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
109.	C-Ring surfaces damaged		100%	3.2	4.5.3
110.	Motor body or bulkhead bonded subassembly seal surfaces damaged		100%	3.2	4.5.3
111.	C-Ring missing or does not seat		100%	3.2	Visual
112.	Oversize pin holes excessive in number		100%	3.2	Visual
113.	Pin holes not clean and free of dirt, rust and contamination		100%	3.2	Visual
114.	Pins and retaining rings not clean and free of dirt, rust and contamination		100%	3.2	Visual
115.	Improper motor body/bulkhead orientation (by scribe mark or alignment pins)		100%	3.2	Visual/Manual
116.	Leak test before pin insertion		100%	3.6	4.5.5
117.	Any retaining pin missing or improperly assembled		100%	3.2	Visual
118.	Leak test after pin insertion		100%	3.6	4.5.5
119.	Inspection records for successful reinspection of CD103 through CD118 (when lapping is required)		100%	3.2	Visual
120.	Forward face to seal surface (when lapping is required)		100%	3.2	Gage
121.	Aft face to seal surface (when lapping is required)		100%	3.2	Gage
122.	Any forward facing steps (360°)		100%	3.2	Gage
123.	Any retaining ring missing or incorrectly assembled		100%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6 (cont)	Rocket Motor/Bulkhead Assembly	3 OF 3	9299058 (see 6.10) 9342458
			NEXT HIGHER ASSEMBLY: M753 PROJ

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
124.	Improper retaining ring gap before pull test (.086 min.)		100%	3.2	Gage
125.	Improper retaining pin or ring extraction force		100%	3.5	4.5.4
126.	Improper retaining ring gap after pull test (.086 min.)		100%	3.2	Gage
127.	Monitoring port ball or monitoring port ball seal surfaces not clean and free of dirt, rust and contamination		100%	3.2	Visual
128.	Corrosive preventive compound missing from monitoring port threads		100%	3.2	Visual
129.	Improper monitoring port setscrew torque or setscrew not below motor surface		100%	3.2	Gage
130.	Corrosion preventive compound missing from retaining ring holes and pins		100%	3.2	Visual
131.	Rocket-off cap visibly cracked or damaged		100%	3.2	Visual
132.	Rocket-off cap torque (recheck)		100%	3.2	Gage
133.	Marking missing, improper or illegible		100%	3.2	Visual
134.	Protective finish improper or missing		100%	3.2	Visual
135.	Poor workmanship		100%	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Packaging				See 5.0
					NEXT HIGHER ASSEMBLY
					N.A.
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101.	Shipping container improper or damaged		100%		Visual
102.	Grommet assembly improper or missing		100%	5.0	Visual
103.	Any container support or cushion missing or improperly assembled		100%	5.0	Visual
104.	Filler material insufficient to provide firm pack		100%	5.0	Visual/Manual
105.	Dessicant improper or missing		100%	5.0	Visual
106.	Data cards or inspection records missing		100%	3.4	Visual
107.	Recorded weight of gold missing from shipping documents		100%	3.2	Visual
Minor					
201.	Poor workmanship		100%	---	Visual
202.	Marking incorrect, missing or illegible		100%	5.0	Visual

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4.4.3 Testing. Testing shall be performed as described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract (see MIL-A-48078 and 6.3 herein).

4.5 Test methods and procedures.

4.5.1 X-ray examination for missing or improperly assembled components (see applicable Rocket Motor Assembly Dwg.). Each assembly shall be subjected to X-ray examination perpendicular to the longitudinal axis at the chalked scribe line in a plane that will clearly define all of the components, including both channels of the delay assembly. Necessary controls will be implemented to assure that the assembly is subject to X-ray in a plane that will include both channels. In all exposures the X-ray beam shall form a 90 degree angle with the longitudinal axis of the motor body. The test shall be performed as specified in 4.5.1.1. Any assembly which shows evidence of missing or improperly assembled components, as listed below, shall be classed as defective and removed from the lot.

a. Delay housing assembly not properly seated in motor body.
b. Shear ring seal not fully expanded in delay and motor body grooves.

(The groove in the motor body shall be completely filled with lead. There shall be no gap between the shear ring seal and the shoulder of the expansion cone.)

c. Rocket-off cap assembled without seal or not fully engaged.
d. Evidence of foreign material present.
e. Any other deviation from the normal assembly condition.

4.5.1.1 Radiographic equipment, operations and procedures. Radiographic equipment, operations, and procedures shall comply with MIL-STD-453. In addition, the sensitivity

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of the X-ray technique shall be 2.0 percent and the photographic density of the film used shall be no less than 1.25 as determined in accordance with 4.5.1.1.2.

4.5.1.1.1 Check of sensitivity of X-ray technique. The sensitivity of X-ray techniques shall be determined by the means of a penetrameter fabricated as specified herein. The penetrameter shall be placed on the side of the motor body away from the film (source side). Observation shall be made of the image of the penetrameter on the X-ray film. The sensitivity of X-ray technique shall be considered satisfactory when the image of the penetrameter including the 2T hole is sharply defined on the X-ray film. Unless otherwise specified, the penetrameter shall be made of steel having an approximate specific gravity of 7.8. The thickness of the penetrameter shall be based on maximum steel thickness and shall be equal to 2.0 percent of the maximum thickness. The penetrameter shall have a hole the diameter of which is twice the thickness of the penetrameter. For further details on construction of the penetrameter see MIL-STD-453. The actual thickness may be determined from a radiograph of the metal parts.

4.5.1.1.2 Check of photographic density of film. The photographic density of the film shall be determined on the film in a region of sound charge plus metal corresponding to the portion of the material which is of the greatest combined thickness to be examined on the negative. To judge densities, it is recommended that standards be used. "American Standards for Diffuse Transmission Density" published by American Standards Association, 70 East 40th Street, New York has been satisfactory as a source of information of film densities. Comparison of the film with standards may be made visually; however, the use of a densitomometer is recommended.

4.5.2 Infrared inspection. Infrared inspection shall be performed with approved test equipment, standards, calibration instructions and procedures.

4.5.3 Seal and seal surface defects. Prior to loading and assembly, motor body, bulkhead and C-Ring sealing surfaces shall be examined with a 30x magnifier for defects. Any sealing surface which exhibits non-circumferential scratches or flaws shall be rejected. Rejected seal surfaces of motor bodies and bulkheads may be reworked by lapping in accordance with dwg. requirements to meet the seal leak rate requirements of 3.6.

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4.5.4 Pin extraction force (see 3.5). Pin and retaining ring extraction force shall be measured with approved test equipment.

4.5.5 Leak rate (see 3.6). The leak rate of the rocket motor/bulkhead joint and pin holes shall be measured with an approved helium leak detector, procedures and enclosure or fixture. The tracer gas shall be helium meeting the requirements of FED-SPEC BB-H-1168 (Grade A).

4.5.6 Delay removal force (see 3.7). The nozzle insert, grain support and aft cushion need not be assembled for this test. The test shall be performed after the delay assembly has been assembled in the motor body and X-rayed for missing or incorrectly assembled components in a normal manner. The test shall be accomplished with approved test fixtures and a suitable forcing mechanism provided with strain gauge or other force measuring apparatus to measure the applied force.

4.5.7 Proving ground test (see 3.8). These tests shall be performed at a Government proving ground with rocket motors assembled on PXR6275-2 Rocket Test Vehicles (see dwg. 9328029) by the acquiring agency. Testing shall be performed with an approved test program request (TPR) provided to the proving ground commander, and witnessed and evaluated by the cognizant PAD element of ARRADCOM (see 6.4). The test vehicle shall be fired at 130°F in the rocket-on mode from an M201 series cannon using an M188A1 propelling charge conditioned at 145°F (Zone 9) with chamber pressure of 44000 $\pm$ 1000 psi. Adequate photographic and instrumentation coverage shall be provided to assure compliance with 3.8 including verification of projectile structural integrity, and ignition and burning of the delay assembly columns. Exterior ballistic range and deflection performance shall also be recorded. In the event of any abnormal behavior other than specified in 3.8 (including failure to perform to M650 interior and exterior ballistic characteristics), an attempt shall be made to recover and examine the rocket motor/bulkhead to establish the cause of failure.

5. PACKAGING

5.1 Packaging, packing and marking. Unless otherwise specified in the contract, assemblies shall be packaged, packed and marked in accordance with the applicable requirements of dwg. 9311169.

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6. NOTES

6.1 Intended use. Rocket Motor/Bulkhead Assemblies covered by this specification are intended for use in Assembly of M753 8 Inch Projectiles and for assembly of DOE test projectiles (see 6.10).

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Clean room approval (see 3.3).

c. First article inspection, when required (see 3.10).

d. Government furnished property, as applicable (see 6.9).

e. Type 6B Assembly, where required (see 6.10).

6.2.2 Data requirements. When this specification is used in an acquistion which incorporates a DD1423 Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR7-104.9 (n) (2) are involved and a DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraph:

Paragraph no.	Data requirements	Applicable DID no.
(a) 3.4	Data cards	DI-L-1410
(b) 3.4	Data feedback (inspection records)	DI-R-1724
(c) 4.4.4	Inspection equipment approval	DI-R-1714 (Tailored)

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.3 Equipment design approval. Submit equipment designs, as required, to Commander U S Army Armament Research and Development Command, ATTN: DRDAR-QAN-I, Dover, New Jersey 07801.

6.4 Ballistic tests. TPR preparation, test firing and test evaluation shall be performed by QEB, NSD, PAD (DRDAR-QAN-Q). A short summary of test results shall be furnished to the Evaluation and Metrology Div., PAD (DRDAR-QAM) within 5 days after completion of the test for inclusion in the first article test report.

6.5 Adhesive approval. Submit adhesives and adhesive test coupons for approval to Commander US Army Armament Research and Development Command, ATTN: DRDAR-LCA-OA, Dover, New Jersey 07801.

6.6 Objective evidence.

6.6.1 Materials. Certification (supported by test data) that materials comply with applicable dwg. and specification requirements.

6.6.2 Components. Inspection records, data cards or other documentary evidence of previous Government acceptance in accordance with applicable dwgs. and specifications.

6.7 Clean room. The clean room shall be approved by Commander, U.S. Army Armament Research and Development Command ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801. As a minimum, the clean room shall be temperature and humidity controlled to assure a controlled temperature of 75 $\pm$ 5°F with a maximum relative humidity of 50 percent. Adequate means shall be provided to monitor and control the humidity and temperature. Personnel shall wear low article shedding type garments and adequate approved means shall be provided to control dirt and loose particles and to dispose of contaminated materials.

6.8 Supersession data. This specification includes the requirements of Purchase Description AR-PD-6, Revision A dated 5 August 1980.

6.9 Government furnished property. The contracting officer should arrange to furnish the property listed in 3.9, as applicable.

6.10 Type 6B, Live Assemblies. Type 6B, Live, Rocket Motor/Bulkhead Assemblies furnished to DOE under dwg. 9347149 for test purposes are equivalent to M753 WR Rocket

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Motor/Bulkhead Assemblies, dwg. 9299058. Inspection for 9347149 assemblies shall be equivalent to the inspection requirements of dwg. 9299058 cited herein.

6.11 Drawings, specifications and publications. Drawings, specifications and publications listed in Section 2 of this specification under the heading U S Army Armament Research and Development Command (ARRADCOM) may also include drawings, specifications and publications prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal documents. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

Custodian:

Army-AR

Preparing activity

Army-AR

(Project 1110-A160)

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